

Sales Data Visualization Dashboard Using Python

Internship Domain :

Pratinik Infotech - Python Internship Project

Submitted By :

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Abstract

This project presents the development **Sales Data Visualization Dashboard** using Python. The dashboard enables businesses to analyze their sales data through interactive charts and filters, providing valuable insights category-wise distribution. The project uses **Streamlit** for the web interface, **Pandas** for data handling, **Plotly** for visualization.

Problem Statement

In today's data-driven world, businesses generate large amounts of data. However, raw data is difficult to interpret without proper visualization. This project addresses that by creating an interactive dashboard where users can upload sales data, filter it, and view meaningful visualizations. This enables quick decision-making and improved insights.

Objectives

1. To upload and process sales data from CSV files.
2. To provide filtering options for date, category, and product.
3. To calculate and display key performance indicators (KPIs).
4. To visualize sales and profit data using multiple chart types.
5. To allow users to download filtered data as CSV.

System Requirements

Hardware: A PC/laptop with minimum 4GB RAM and 1GHz processor.

Software: Python 3.9+, Streamlit, Pandas, Plotly, and an IDE like VS Code.

Methodology

- 1. Data Input:** Upload CSV file or use default sample data.
- 2. Data Preprocessing:** Parse dates, filter data based on user input.
- 3. KPI Calculation:** Compute Total Sales, Average Sales, and Total Profit.
- 4. Visualization:** Generate charts (Line, Bar, Pie, Scatter) using Plotly.
- 5. Data Export:** Provide download button for filtered data as CSV.

Features

- Upload sales data from CSV.
- Apply filters by date, category, and product.
- View KPIs: Total Sales, Average Sales, Total Profit.
- Multiple visualizations for better insights.
- Download filtered data in CSV format.

Output

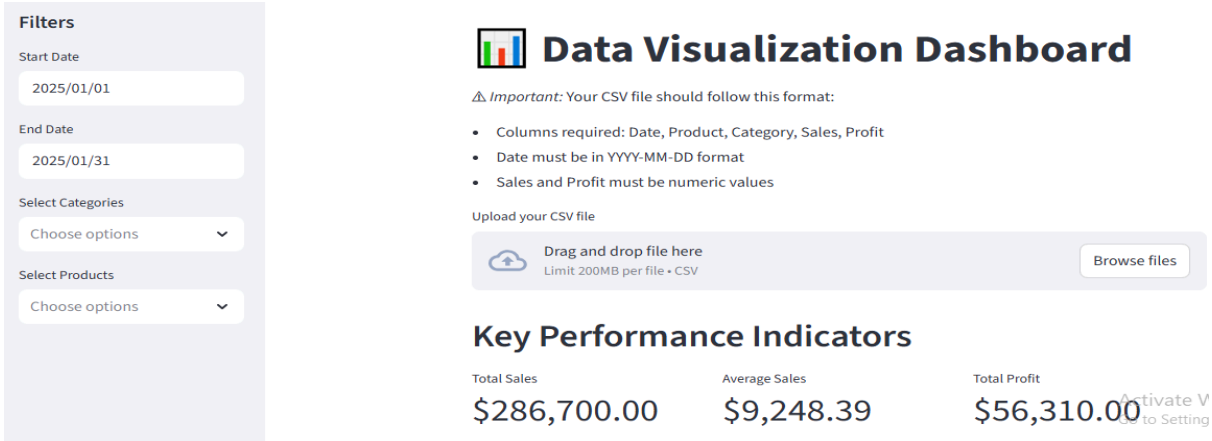


Fig 1: Dashboard Homepage with KPI(Key Performance Indicators)

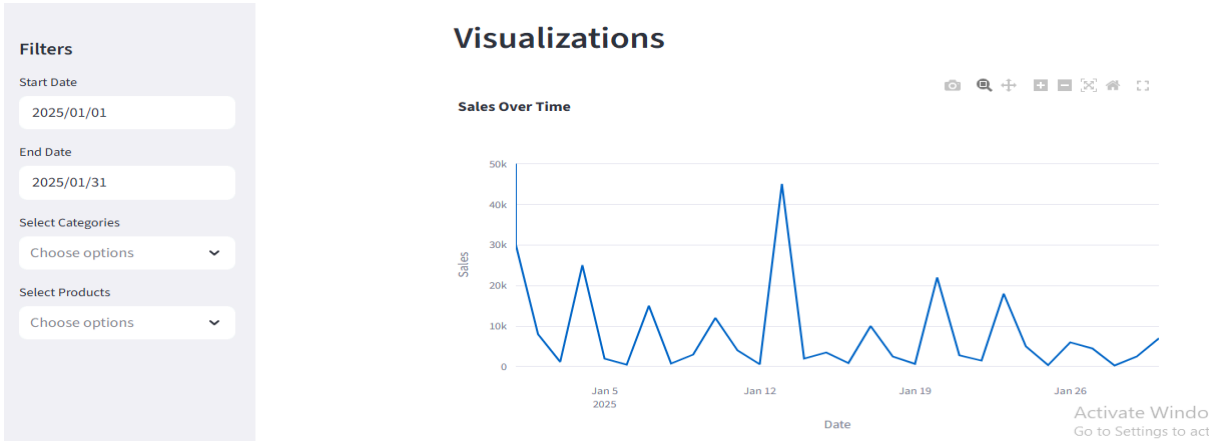


Fig 2: Line Chart showing Sales Trend

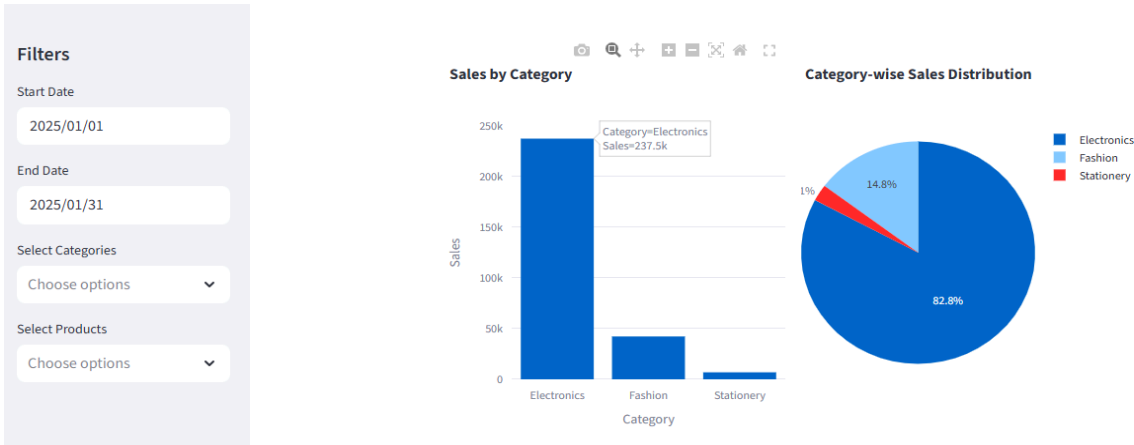


Fig 3: Pie chart and Bar graph of product contribution

Filters

Start Date

2025/01/01

End Date

2025/01/16

Select Categories

Choose options

Select Products

Choose options

Sales vs Profit

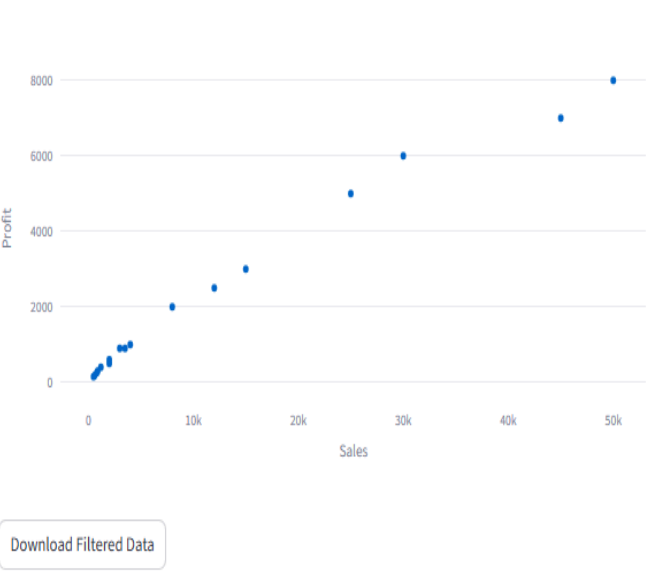


Fig 4: Scatter plot showing Quantity vs Sales

Filters

Start Date

2025/01/01

End Date

2025/01/31

Select Categories

Electronics

Fashion

Stationery

Key Performance Indicators

Total Sales	Average Sales	Total Profit
\$237,500.00	\$19,791.67	\$43,500.00

Visualizations

Sales Over Time



Fig 5: Dashboard after applying Electronics category filter

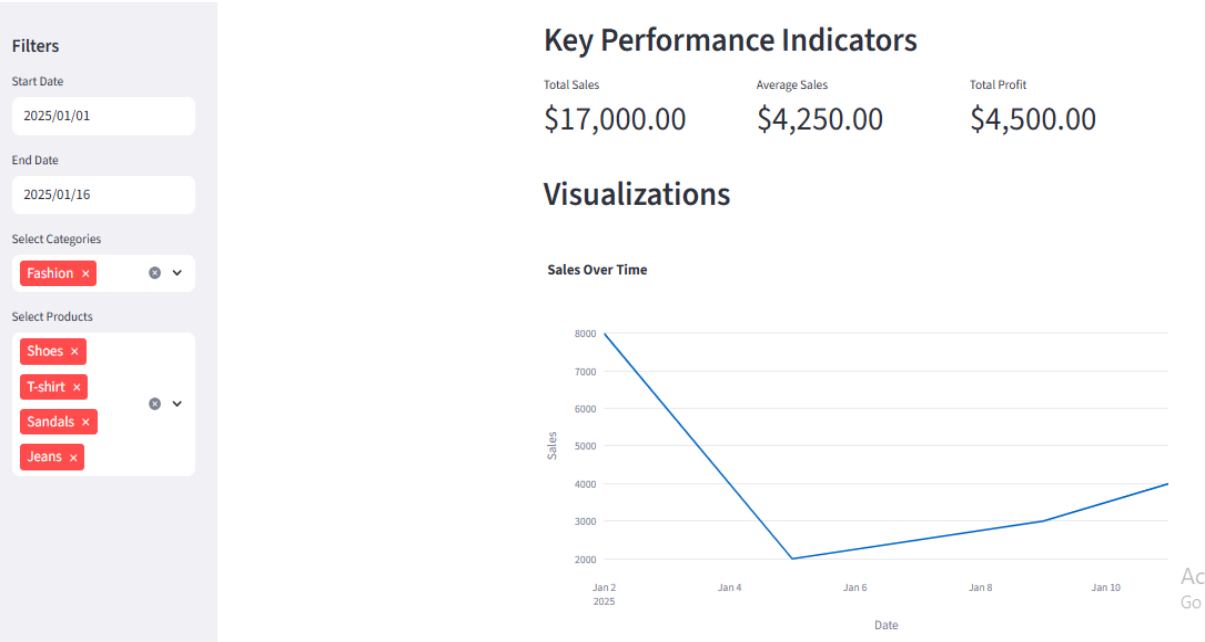


Fig 6: Dashboard after applying Fashion category and select products filter

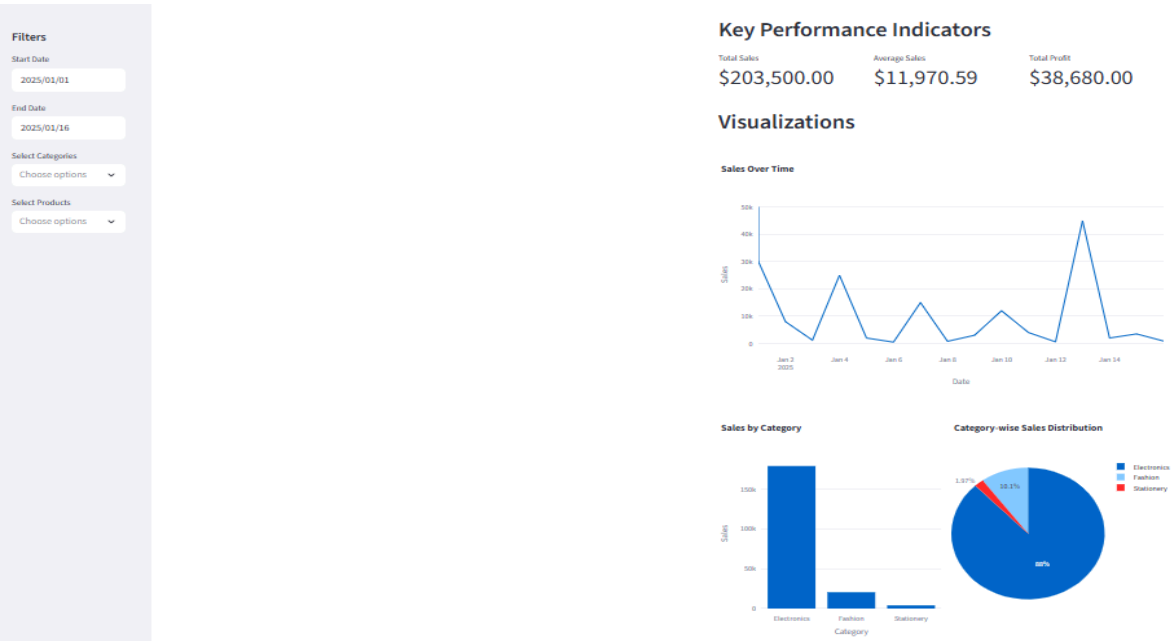


Fig 7: Dashboard after selecting Date filter(01-16 JANUARY)

Conclusion

The project successfully demonstrates the creation of an interactive sales data visualization dashboard. Users can easily explore sales data, identify trends, and analyze category-wise performance. The project highlights the use of Python for data analysis and visualization in real-world applications.

Future Scope

- Add machine learning predictions for sales trends.
- Integrate authentication for multiple users.
- Deploy on cloud platforms like Streamlit Cloud or Heroku.

References

- **Streamlit Documentation:**
<https://docs.streamlit.io/>
- **Pandas Documentation:**
<https://pandas.pydata.org/>
- **Plotly Documentation:**
<https://plotly.com/python/>

